Does Instance Segmentation Fix Person–PPE Association?

A controlled refutation, and the geometry behind it

Project 7 · Workplace-safety compliance monitoring

SH17 · YOLOv8 · MobileSAM

All figures generated from `outputs/*.json`

1. The question, and why it is not obvious

A workplace-safety monitor has to answer a question about *people*, not about pixels: *is this worker wearing head protection?* Object detection alone cannot answer it. A frame containing one `person` box, one `head` box and one `helmet` box becomes ambiguous the instant a second person walks in — the helmet must be **assigned** to somebody, and the assignment rule, not the detector, decides the verdict.

The error is asymmetric. Scoring a non-compliant worker as compliant leaves a person unprotected on a site; the reverse merely triggers a check that costs a supervisor a minute. The headline metric throughout this book is therefore **per-person non-compliance recall**, and "undetermined" is counted as a **miss**: a monitor that cannot tell protects nobody.

The intuition that motivates the project is genuinely appealing. When two workers overlap in frame, their bounding *boxes* may both contain the same helmet, so a box-overlap test is close to a coin flip in exactly the crowded case that matters. Their instance *masks*, however, are disjoint. Segmentation looks like the principled fix.

> **Hypothesis, fixed before any training run.** Mask-containment association yields higher per-person non-compliance recall than box-containment association, and the margin **increases** with the number of persons per frame.

Both limbs are falsifiable. This book shows both are false, and — more usefully — measures exactly which assumption breaks.

> **Result.** With detection removed entirely (ground-truth boxes, real segmentation), mask containment scores **0.7991** non-compliance recall against **0.9329** for box containment: Δ = **-0.1338**, 95% CI [-0.1521, -0.1157], P(Δ>0) = 0.000 over 700 scenes and 3,029 persons. The margin does not grow with crowding. Root cause: for **26.5%** of helmets, the wearer's own mask covers less than half the helmet box.

1.1 Decision rule, fixed in advance

The quantity that decides the hypothesis is the paired difference in non-compliance recall, with a 95% confidence interval from a bootstrap that resamples *composites* rather than persons, plus the per-stratum profile across crowding bands 1, 2–3, 4–6 and 7+. A flat profile refutes the second limb even if the first were to hold. Fixing this before running anything is what stops the metric being chosen after the fact to suit the result.

2. Dataset and splits

The corpus is **SH17**, a workplace-safety detection dataset of 8,099 images and 75,994 instances across 17 classes. It was selected for a property most PPE datasets do not have: it annotates `person`, `head`, `helmet` and `safety-vest` as **independent boxes** and never records which head belongs to which person.

That absence is the entire reason it is usable here. A dataset that ships per-person compliance verdicts has already solved the association problem inside its labels, and any experiment run on it measures label quality rather than the mechanism. The cost of that choice is that there is no association ground truth to score against either — addressed in Chapter 4.

Class map recovered, not assumed

The class index → name mapping was recovered empirically by aligning YOLO line order against VOC object order across 400 random images, achieving purity 1.000 on all 17 classes. It was not copied from the paper. A silently wrong helmet index produces a fully functional pipeline that computes a meaningless number, and nothing errors.

Split protocol

```
SH17 train (6,479) ─┬─ det_train  5,831   detector fitting
                    └─ det_val      648   checkpoint selection
SH17 val   (1,620) ─── HELD OUT           composite sources + test mAP
```

The authors' own train/val boundary is never crossed. Every person appearing in the evaluation set comes from an image the detector has never seen.

A narrowing forced by the data

The brief specified a compliance verdict of helmet *and* vest. Counting the held-out single-person pool before any arm was scored: 712 images with head-and-no-helmet, 35 with head-and-helmet, and **14** containing a safety vest. Fourteen source images cannot support a headline metric, so **compliance is head-protection only**. The vest remains a detector class — it is nearly free and its mAP is reportable — but it is not in the verdict. This is a narrowing forced by counting, not a change of question after seeing results.

3. The three arms

Every arm consumes **one** detection pass per image. Only the association step differs, so no observed difference can be attributed to detector noise. Writing p for a person index and j for a PPE box index, all three arms are the same argmax rule under a different region and score:

$$\mathrm{owner}(j) = \arg\max_p \; s(j,p), \qquad \text{assign iff } \max_p s(j,p) \geq \tau$$

with the arms differing only in s :

Arm	Person region	Score s(j,p)	τ
A0	bounding box	IoU(b_j, b_p)	0.30
A	bounding **box**	|b_j ∩ b_p| / |b_j| (containment)	0.50
B	instance **mask**	|b_j ∩ M_p| / |b_j| (containment)	0.50

Intersection-over-union and containment are different questions. IoU asks how much two regions agree; containment asks what fraction of the *small* region lies inside the large one:

$$\mathrm{IoU}(a,b) = \frac{|a \cap b|}{|a \cup b|}, \qquad \mathrm{cont}(a,b) = \frac{|a \cap b|}{|a|}$$

The compliance rule applied identically to every arm's assignment:

$$v(p) = \begin{cases} \texttt{non-compliant} & \text{head assigned to p, no helmet assigned to p} \\ \texttt{compliant} & \text{helmet assigned to p} \\ \texttt{undetermined} & \text{neither assigned} \end{cases}$$

> **This derivation is an assumption, not a dataset fact** — and it is precisely the assumption the two arms disagree about. It is stated here rather than buried in a config file. A person with no evidence at all is left undetermined rather than scored compliant: scoring absence as compliance is the exact failure mode that makes a safety monitor useless.

3.1 Why ARM A0 exists, and why ARM A is the honest baseline

The original scaffold specified ARM A0: assign each PPE box to the person box it overlaps most, above IoU ≥ 0.30. That baseline is broken, and not in an interesting way. A helmet box occupies roughly 1–2% of a person box's area, so

$$\mathrm{IoU}(b_{helmet}, b_{person}) \approx \frac{|b_{helmet}|}{|b_{person}|} \approx 0.015$$

a factor of twenty below its own threshold. It can never fire. Measured on the oracle run: **0.0227** non-compliance recall at an undetermined rate of **98.5%**.

Had that been left as the baseline, ARM B would have "won" by roughly sixty points against an arm that never assigned anything — and no code would have errored, no test would have failed, and the write-up would have reported a spectacular fake result. The fair baseline is the rule practitioners actually use for small-object-to-container assignment: containment with argmax.

> **A0 is retained and reported, not deleted.** It costs nothing (it reuses the same detections) and it is the evidence that ARM A was chosen to be fair rather than chosen to lose. A reviewer asking "did you just pick a weak baseline?" is answered by a row in a table instead of a promise.

4. Ground truth: synthetic composites

SH17 records no person↔PPE link, so there is nothing to score association against. Four options were considered:

Option	Ground truth	Verdict
Score against box-IoU-derived links	Circular — defines ARM A correct	**Reject**
Evaluate only on single-person images	Unambiguous, but zero crowding	**Reject** — crowding is the claim
Hand-label a crowded subset	Real, but slow and small	Keep as external check
Synthetic composites	**Exact, by construction**	**Primary**

Take held-out images containing exactly *one* person — in those, every PPE box provably belongs to that person — and composite k of them into a single crowded scene, carrying the known links through. This yields exact ground truth at any crowding level and turns crowding into a *swept* variable rather than whatever distribution the corpus happens to contain.

Built: **700 composites, 3,029 persons** (1,981 non-compliant), mean 4.33 persons per scene, maximum 9.

4.1 Two generator choices that would have decided the result

Trap 1 — a "better looking" paste rigs the outcome. The natural instinct is to segment each source person with SAM and paste the silhouette so the scene looks realistic. That is a bug: a silhouette pasted onto a new background leaves a sharp figure/ground boundary that SAM can re-segment almost trivially at evaluation time. The mask arm would win because the generator drew its answer into the image. Rectangular crops carry their own background, so the segmentation step has to do real work — uglier, and conservative toward the project's own hypothesis.

The general form is worth stating: *if the generator uses model M to build the data, any arm using M gets a free signal.*

Trap 2 — a fixed canvas confounds the independent variable. The first implementation kept a 1280 px canvas and shrank everyone to fit, so 9-person scenes had persons at roughly half the size of 2-person scenes. The claim under test is that the margin *grows with crowding* — but crowding and object scale were now moving together, so any trend would have been unattributable. Fixed by holding person height constant and widening the canvas with the crowd.

Two further problems were invisible in the metrics and obvious on one look at a rendered composite: SH17's zero-person images are object close-ups (a jar, a hand) which pasted sharply behind workers looked absurd, and persons were being clipped by the canvas bottom, producing unusable ground-truth boxes. **Render a handful with ground truth overlaid before generating the full set** — every time.

5. Estimators

Non-compliance recall

Over all ground-truth non-compliant persons, including those the detector never found:

$$R_{strict} = \frac{\#\{p:\ v_{gt}(p) = \mathrm{NC}\ \wedge\ v_{pred}(p) = \mathrm{NC}\}}{\#\{p:\ v_{gt}(p) = \mathrm{NC}\}}$$

A *conditioned* variant restricted to persons the detector actually matched is also reported; it isolates the association step. Because both arms consume identical detections, the detector's contribution cancels from the A→B difference under either definition.

Compliant F1 — the guard metric

An arm that maximises non-compliance recall by declaring everyone non-compliant scores 1.000 and is worthless. F1 on the compliant class is reported alongside every recall figure to make that degenerate strategy visible.

Paired bootstrap over composites

Persons within a scene are correlated through the scene they share, so resampling persons independently would understate variance. The bootstrap resamples *composites* with replacement, $B = 2000$ replicates, and both arms see the same resampled scenes in each replicate — so the detector's contribution cancels within the replicate rather than merely on average:

$$\hat{\Delta} = R_B - R_A, \qquad \mathrm{CI}_{95} = [\ q_{2.5}(\Delta^*),\ q_{97.5}(\Delta^*)\ ]$$

6. The oracle ablation — the decisive experiment

The full pipeline feeds both arms the same detections, which controls for detector quality but does not remove it: a missed helmet hurts both arms and compresses the measurable gap. Substituting ground-truth boxes removes detection from the loop entirely and asks the mechanism question in its purest form. MobileSAM still segments the real composite image and has no knowledge of how it was built.

> This reading was **committed to in advance**, in the `src/oracle.py` docstring: *if ARM B shows no advantage here, it cannot show one with a real detector either, and the hypothesis is dead regardless of how well the detector is trained.* Pre-registering the interpretation is what stops an inconvenient result being re-read as "we just need a better detector".

Arm	nc-recall	helmet assoc. acc.	undetermined	compliant F1
A0 — box IoU @0.30	0.0227	0.0000	0.9848	0.0000
A — box containment @0.50	**0.9329**	**0.8986**	0.0409	**0.9332**
B — mask containment @0.50	0.7991	0.7181	0.1799	0.8561

B − A = -0.1338, 95% CI [-0.1521, -0.1157], P(Δ>0) = 0.000

The directional claim

The second limb of the hypothesis was that the margin grows with crowding. By stratum:

1 person	2–3	4–6	7+
-0.0185	-0.1467	-0.1451	-0.1217

Flat, and negative throughout. Both limbs of the hypothesis fail simultaneously, with detection removed from the loop.

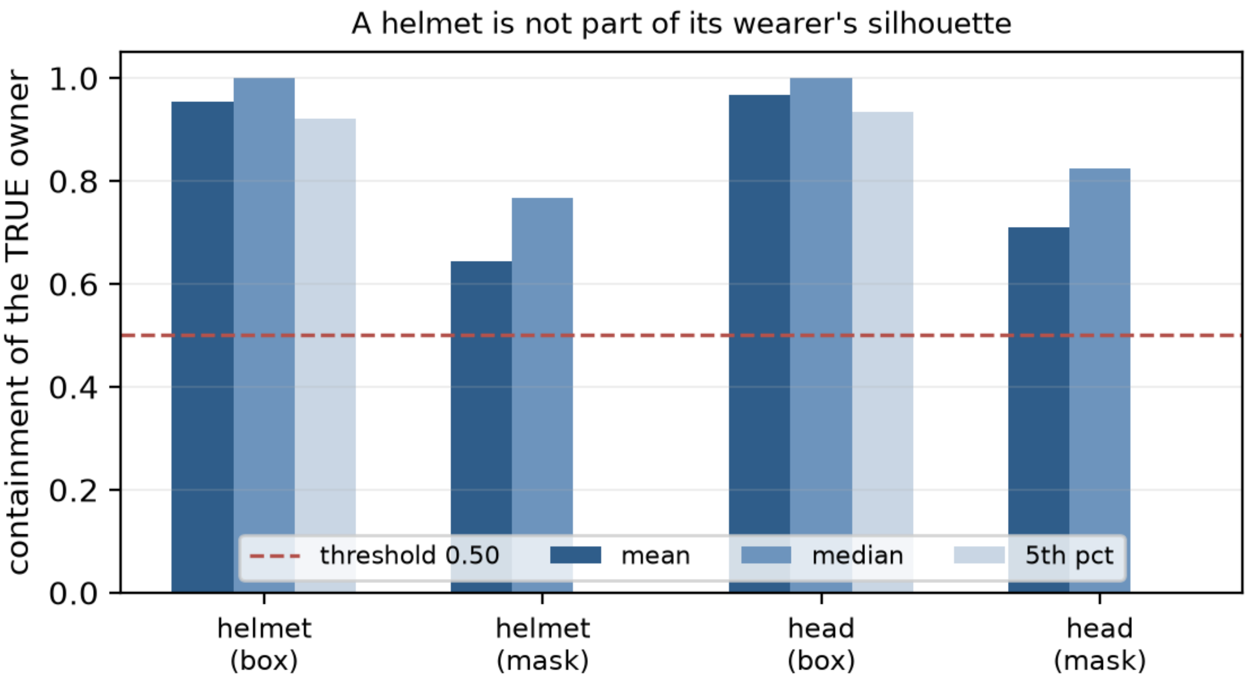
7. Why masks lose — the containment distribution

Before accepting a refutation it is worth asking whether the measurement was fair. A person's segmentation mask is very nearly a *subset* of their bounding box, so

$$\mathrm{cont}(b_j, M_p) \;\le\; \mathrm{cont}(b_j, b_p) \qquad \text{for every } j, p$$

by construction. The two arms are therefore **not scored on the same scale**, which is a problem addressed fully in Chapter 8. First, the direct diagnostic: score every ground-truth (PPE box, *known* owner) pair under both rules, which requires no thresholds and no assignment at all.

	mean	median	p05	fraction < 0.50	n
helmet → owner **box**	0.954	1.000	0.920	4.1%	1,114
helmet → owner **mask**	0.643	0.767	**0.000**	**26.5%**	1,114
head → owner **box**	0.968	1.000	0.934	2.8%	3,125
head → owner **mask**	0.709	0.823	0.000	20.6%	3,125



Containment of PPE boxes within their known owner's region. The box rule sits at ceiling; the mask rule has a fifth percentile of zero.

> **This is the whole finding in one table.** For 26.5% of helmets the wearer's own mask covers less than half the helmet box, and the 5th percentile is 0.000 — a substantial tail of helmets falls *entirely outside* the mask of the person wearing them.

This is not a SAM failure. SAM is doing exactly its job: prompted with the person's box, it segments *the person*. A hard hat is an object worn **on top of** a person and is not part of their silhouette. ARM B's elevated undetermined rate (18.0% against 4.1%) is precisely this: the true link exists and the mask test rejects it.

The bounding box has no such problem *because* it is crude, and crudeness is what this task wants. Segmentation answers "which pixels are this person". Association needs "which region is associated with this person". Worn equipment falls in the gap between those two questions.

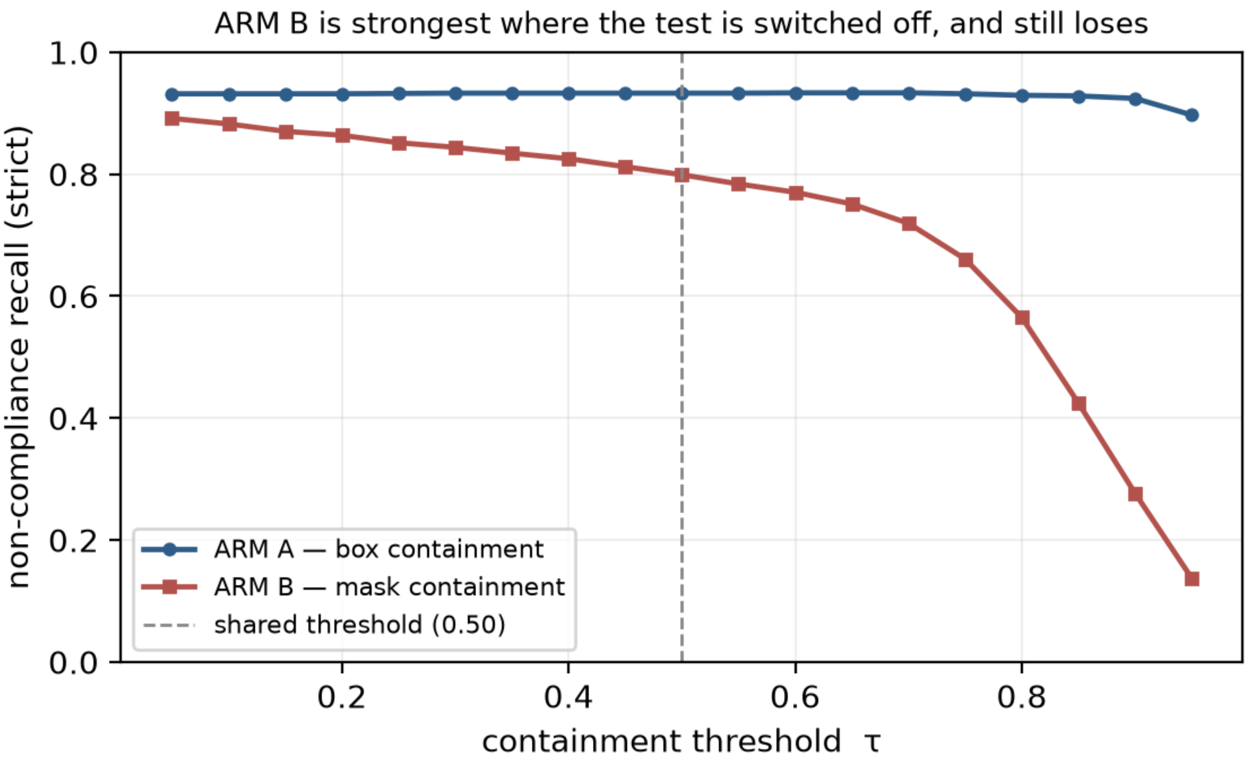
8. Robustness: the threshold sweep

The shared threshold was adopted for a good reason — tuning each arm separately confounds the mechanism with threshold search — and it carries the hidden confound established in Chapter 7. "ARM B loses at 0.50" had two incompatible readings: masks are the worse mechanism, or 0.50 is the wrong operating point for a quantity whose range is compressed. Opposite conclusions, same number. The experiment cannot end at a single threshold.

Both arms were swept over $\tau \in [0.05, 0.95]$ and compared at each arm's own best operating point, where "best" is highest non-compliance recall with ties broken on compliant F1 so a degenerate all-non-compliant point cannot win:

	best τ	nc-recall	compliant F1
ARM A (box)	0.60	**0.9334**	0.9341
ARM B (mask)	0.05	0.8915	0.9216

Best-vs-best B − A = -0.0419, 95% CI [-0.0567, -0.0283], P(Δ>0) = 0.000



ARM A is effectively threshold-independent; ARM B improves monotonically as the test is relaxed and peaks where it is nearly disabled.

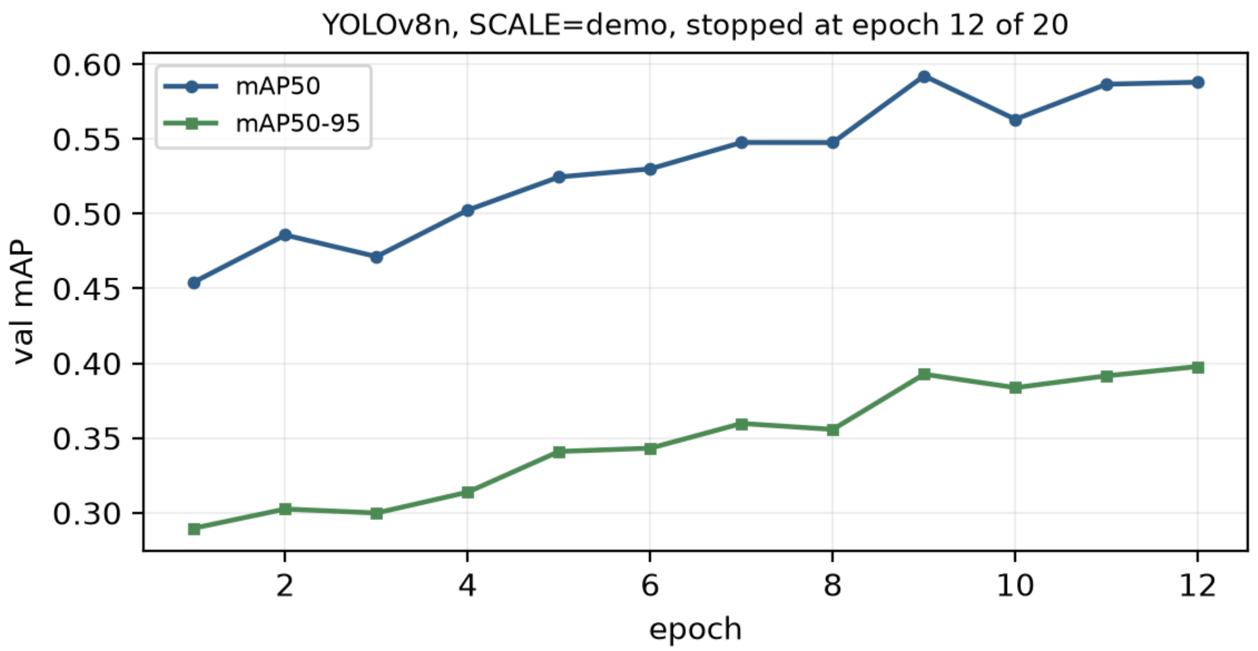
ARM A is flat across the entire sweep because box containment on a true pair is ≈1.0. ARM B improves monotonically as τ falls and peaks at 0.05 — at its strongest when the containment test is switched off almost entirely — and still loses.

> This procedure selects each arm's threshold on the same data it is scored on, which is optimistic for *both* arms and therefore **generous to the arm being refuted**. The refutation survives that generosity.

> Roughly two thirds of the shared-threshold gap (-0.1338 → -0.0419) was the threshold artifact; the remaining third is real and its confidence interval excludes zero.

9. The same experiment with a real detector

The oracle settles the mechanism. This run says what survives contact with imperfect perception. A YOLOv8n detector was trained on the SH17 train split (`SCALE=demo` , 640 px) and its detections shared by all three arms.



Detector quality

Held-out SH17 val (1,620 images, 5,412 instances): **mAP50-95 = 0.4321**, mAP50 = 0.630. Per class: person 0.634, head 0.642, helmet 0.289, safety-vest 0.164. The small classes are the hard ones, which is the expected profile for this corpus — 39.7k of SH17's 76k annotations are under 1% of image area — and it explains why absolute recall below sits well under the oracle's.

Arm	nc-recall strict	nc-recall conditioned	helmet assoc.	undetermined	compliant F1
A0 box IoU	0.0288	0.0391	0.0014	0.9776	0.0019
A box containment	**0.6224**	**0.8463**	**0.8555**	0.3371	**0.6718**
B mask containment	0.6058	0.8236	0.8289	0.3569	0.6554

Δ_{strict} = **-0.0167** [-0.0244, -0.0088] · $\Delta_{\text{conditioned}}$ = **-0.0226** [-0.0332, -0.0119], both P(Δ>0) = 0.000

The detector matched 2,219 of 3,029 ground-truth persons, so absolute recall falls from 0.9329 to 0.6224 and the A→B gap compresses from -0.1338 to -0.0167. Both effects are expected and neither changes the sign: detector misses are identical across arms by construction, so they add a common loss that dilutes the difference without biasing it.

On the truncated schedule. Training was stopped at epoch 12 of the configured 20. The headline finding uses no detector, and the paired design means detector quality can compress the delta but not flip its sign — which is exactly what was observed. The honest cost is that mAP50-95 = 0.432 understates what the full schedule would reach, since `close_mosaic` activates at epoch 13. Detector quality is reported context here, not the claim.

10. Cost, deployment and monitoring

The segmentation pass is not free

Stage	p50	p95
YOLOv8n detection	24.3 ms	39.7 ms
MobileSAM segmentation (ARM B only)	285.9 ms	404.7 ms

Measured over 700 detection calls and 695 segmentation calls during the evaluation run. On the serving path the end-to-end figures were ~611–727 ms for `arm=mask` against ~37 ms for `arm=box` — roughly **17×** the cost. The arm under test is not merely less accurate; it is substantially more expensive, so there is no operating regime in this evaluation where it is the right choice.

Serving

A FastAPI service exposes `/analyze` with the association arm selectable per request, so a deployment can be A/B'd against the same baseline this book reports rather than silently shipping one arm and hoping. A Dockerfile is authored for CPU inference; the trained checkpoint is mounted at run time rather than baked into the image layer.

Label-free drift monitoring

A deployed safety monitor never receives labels, so drift detection has to be behavioural. Three signals are computed over a rolling window and are all label-free: persons-per-frame distribution (is the site more crowded than anything the system was evaluated on?), the **undetermined rate** (are heads failing to associate?), and mean detector confidence. The undetermined rate is the most informative of the three, because it rises before accuracy visibly degrades — and, as Chapter 7 showed, it is the exact quantity that separates the two arms.

11. What the finding is, and what it is not

Inference

The hypothesis is refuted on both limbs, with the mechanism identified rather than merely observed:

1. Mask-containment association does not beat box-containment association. It is 13.4 points worse at the shared threshold and 4.2 points worse at each arm's own optimum, both with confidence intervals excluding zero.
2. The margin does not grow with crowding, under either threshold regime.
3. The cause is geometric and measured: worn PPE lies outside the wearer's segmentation mask for 26.5% of helmets.

Assumptions the inference depends on

The derived compliance rule (head-without-helmet ⇒ non-compliant) is correct — and it is the arms' point of disagreement. Composited scenes are representative of crowding as a *geometric* phenomenon; they are not photorealistic, but both arms see identical pixels and identical detections, so a compositing artifact cannot manufacture a difference *between* arms. MobileSAM prompted with the person box is representative of instance segmentation for this task.

What this does not establish

It does not show segmentation is useless for safety monitoring — only that *mask containment of a PPE box within a person mask* is a worse association rule than box containment. The headline comes from synthetic composites; a hand-labelled real-image check on ~200 genuinely crowded SH17 images is the external-validity step and has not been done, so the claim is "holds on controlled synthetic crowding", not "holds in the field". The 35 distinct compliant source persons are the binding constraint on absolute values, though the paired design protects the delta.

What would make segmentation win

Two variants follow directly from the diagnostic and are stated as future work rather than folded in after the fact, because each changes the mechanism under test: **dilate** the person mask by roughly the scale of the worn item before testing containment, or segment the **helmet** and test mask-to-mask adjacency rather than containment-within-person. The containment distribution predicts both would help; neither is evidenced here.

Generalisation

The failure is not specific to helmets. Any "associate a small worn or carried object with its owner" problem — hi-vis vests, tools, badges, handheld devices — has the same geometry: the object is adjacent to the person, not inside them. The lesson transfers: *when the relation you need is adjacency, a tighter region is a worse instrument.*

12. Research anchors

Kirillov et al. (2023), Segment Anything. The promptable-segmentation formulation ARM B depends on. Its key property was verified before building on it: box-prompted masks for two *overlapping* person boxes come back disjoint. That property does hold — the mechanism had something to exploit, and it still lost for an unrelated reason.

Zhang et al. (2023), MobileSAM. The distilled variant used here, chosen so the segmentation pass is affordable in a per-frame safety monitor. Chapter 10 reports what it actually cost rather than assuming it was negligible.

Ahmad et al. (2024), SH17. The corpus, selected specifically because it annotates person and PPE as independent boxes and records no association — so the question under test is not pre-solved by the labels.

Jocher et al., YOLOv8 / Ultralytics. The shared detector, used identically by every arm so detection quality cannot enter the comparison.

Efron & Tibshirani (1993), An Introduction to the Bootstrap. The paired cluster bootstrap used for every confidence interval; composites are the resampling unit because persons within a scene are not independent.